

20XW87 – DATA MINING PROJECT LAB

USER ENGAGEMENT ANALYSIS

PROBLEM STATEMENT

In the age of social media and digital marketing, understanding and predicting user engagement patterns has become vital for content creators, brands, and marketers. Despite having access to metrics such as likes, comments, shares, and impressions, there is often a lack of understanding of *how* and *when* users engage with content, and *why* certain posts perform better than others. This project addresses the need to uncover hidden behavioural patterns in user interactions through sequential pattern mining and predictive modelling.

PURPOSE

The primary goal of this project is to:

- Analyse and predict user engagement behaviour on social media platforms using advanced data mining techniques.
- Discover frequent sequential patterns in user actions to provide insights for optimizing content strategies.
- Cluster users and content based on engagement to better target audiences.
- Predict future engagement actions using probabilistic models (e.g., Markov Chains).
- Enhance decision-making for marketers and content creators using data-driven engagement trends.

SCOPE

This project involves:

- Data Collection
Web scraping post and engagement data using BeautifulSoup.
- Data Analysis & Mining
Applying techniques such as:
 - Apriori – Identify frequent itemsets of user interactions.
 - GSP, PrefixSpan, SPADE – Discover frequent sequences in time-ordered engagement data.
 - Markov Chains – Predict the next probable user action.
 - Clustering – Segment users or posts based on behaviour and engagement levels using algorithms like K-Means, DBSCAN, Hierarchical Clustering.

- Feature Engineering & Sentiment Analysis
Including:
 - Post metadata (Platform, Type, Timestamp, Content)
 - Engagement metrics (Likes, Comments, Shares, Reach)
 - Audience demographics (Age, Gender, Location, Interests)
 - Campaign IDs & Sentiment labels (Positive/Neutral/Negative)
- Evaluation
Performance of clustering and predictive models using:
 - Silhouette Score
 - Davis-Bouldin Index
 - Adjusted Rand Index
- Insights Extraction
Derive actionable insights for:
 - Best content types and posting times
 - Target audience segmentation
 - Churn prediction and user retention strategies
- Deployment
Interactive dashboards and visualization using:
 - Streamlit / Flask
 - Google Colab for development
 - Data stored in ChromaDB

TECHNOLOGY STACK

COMPONENT	TOOLS AND LIBRARIES
Language	Python
Data Collection	BeautifulSoup
Data Mining	MLxtend, PyFIM, PrefixSpan, SPADE, Scikit-Fuzzy, PySpark
Clustering & Classification	Scikit-learn, Imbalanced-learn
Sentiment Analysis	TextBlob / Custom NLP Pipeline
Visualization & Deployment	Streamlit, Flask
Development Environment	Google Colab, Jupyter Notebook

COMPONENT	TOOLS AND LIBRARIES
Data Storage	Chroma DB

KEY PROCESS AND ALGORITHMS

Apriori

- Detects frequent engagement actions (e.g., Like → Share).
- Helps in understanding which combinations of interactions lead to higher engagement.

GSP (Generalized Sequential Pattern)

- Reveals temporal sequences of engagement.
- Example: Morning post → Likes → Shares.

PrefixSpan

- Mines frequent sub-sequences without candidate generation.
- Provides insights into user interaction flows.

SPADE

- Vertical data format mining for faster pattern discovery.
- Excellent for trend identification and churn analysis.

Markov Chains

- Predicts user's next interaction based on transition probabilities.
- Useful in recommendation systems and engagement optimization.

Clustering

- Groups posts/users into engagement-based segments (e.g., high, medium, low).
- Algorithms used: K-Means, DBSCAN, Hierarchical.
- Evaluation via Silhouette, DB Index, ARI.

SAMPLE PATTERNS AND INSIGHTS

- $\langle \{\text{Video Post}\} \rightarrow \{\text{High Likes}\} \rightarrow \{\text{More Comments}\} \rangle$
Insight: Video content drives stronger engagement.
- $\langle \{\text{Morning Post}\} \rightarrow \{\text{More Shares}\} \rightarrow \{\text{High Engagement Rate}\} \rangle$
Insight: Morning is a prime time for posting.

- $\langle \{ \text{Comment} \} \rightarrow \{ \text{Share} \} \rangle$ (Confidence = 70%)
Insight: Commenting encourages sharing. Promote discussions!
- $\langle \{ \text{Negative Comment} \} \rightarrow \{ \text{Drop in Engagement} \} \rangle$
Insight: Negative sentiment deters interaction; moderation needed.

CONTRIBUTORS

21PW06 – HARITHRA R

21PW08 – JENISA MERLIN D

FUTURE SCOPE

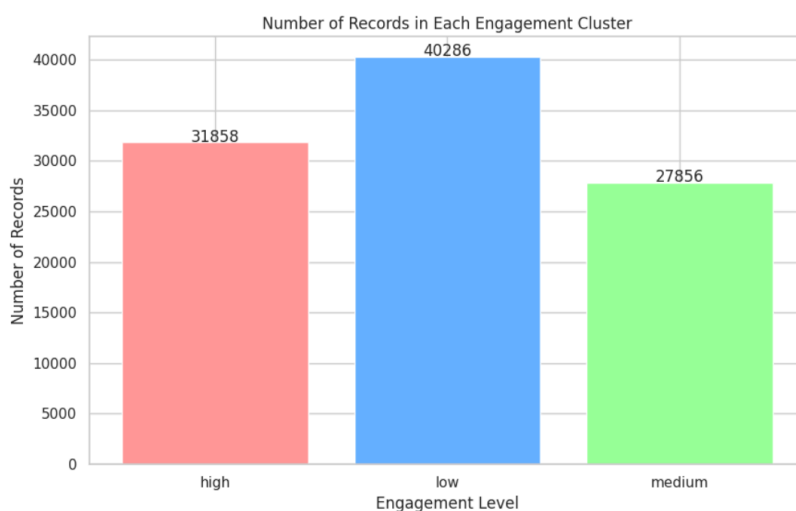
- Integration with real-time social media APIs.
- Dashboard for live engagement trend monitoring.
- Incorporation of deep learning models for better behavioural predictions.
- Integration with CRM systems for campaign tracking.

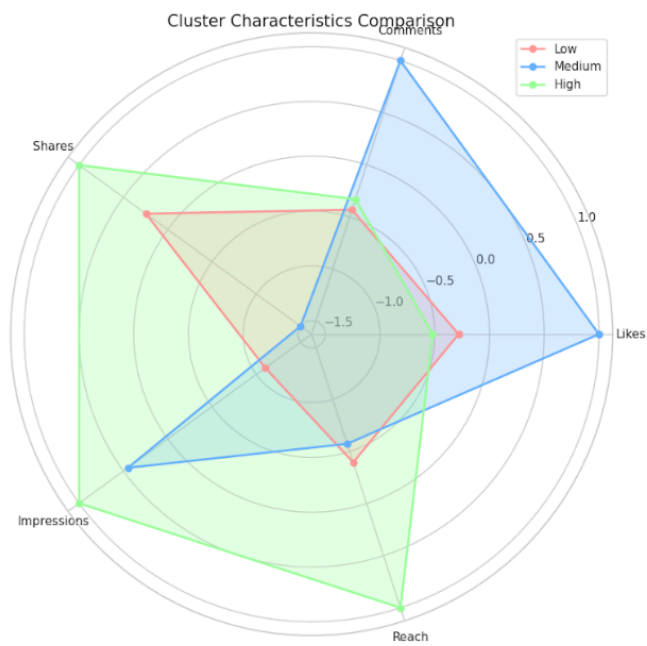
GITHUB LINK:

https://github.com/HarithraR/User_Engagement_Analysis

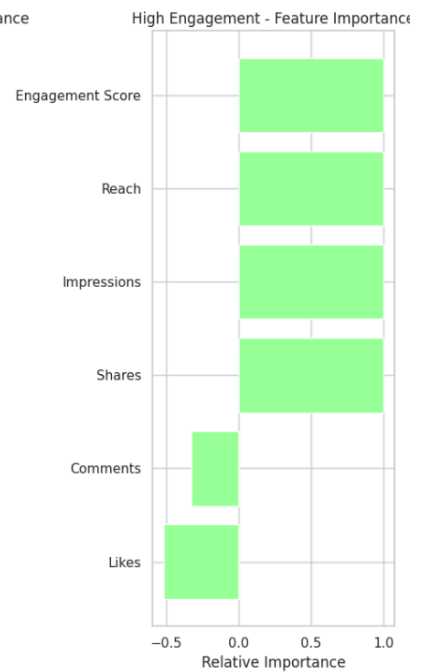
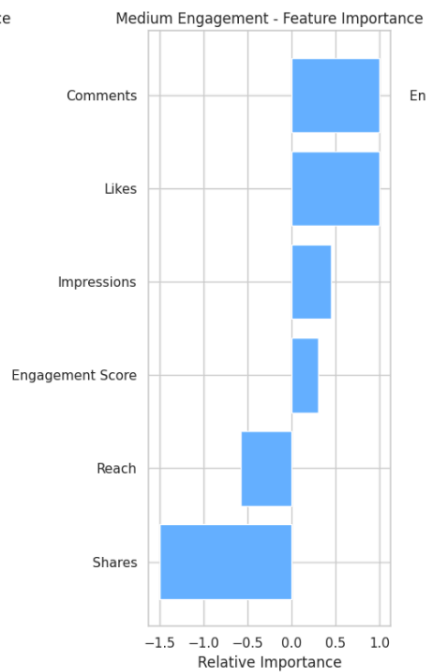
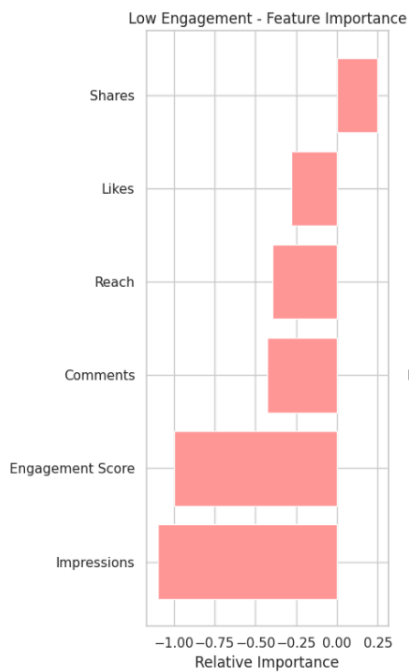
RESULTS:

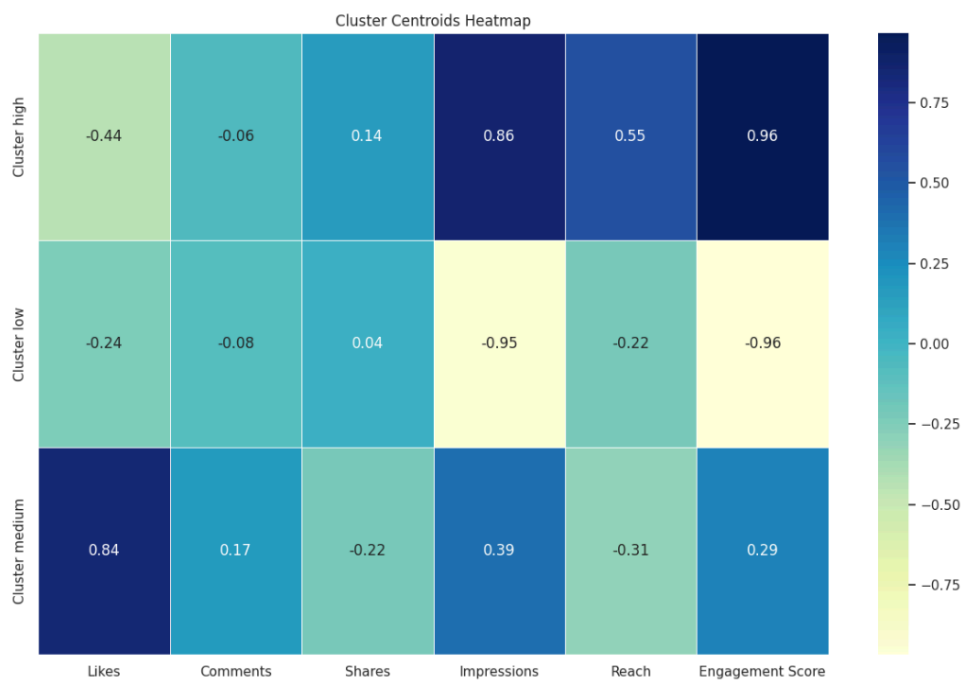
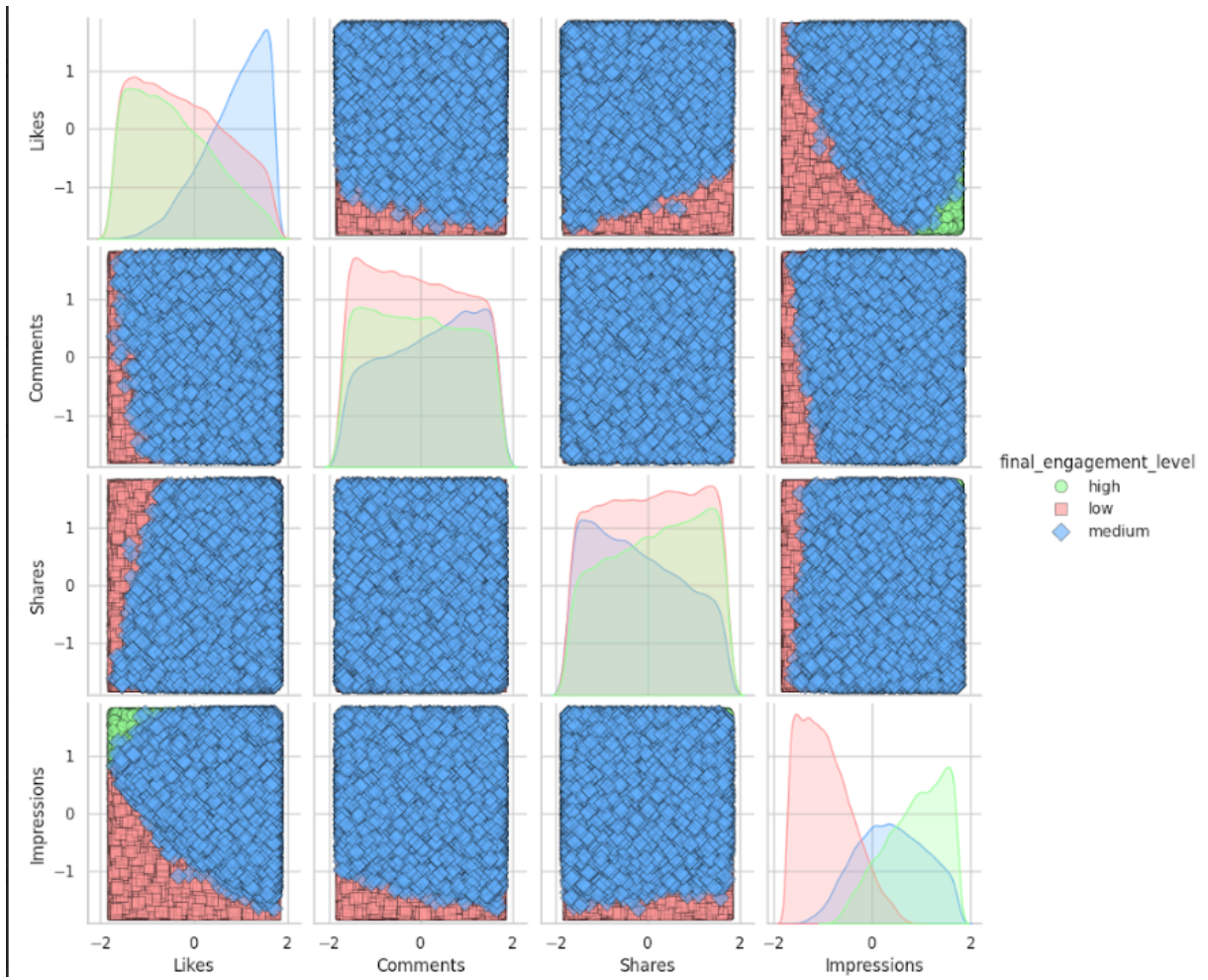
Incremental Clustering:



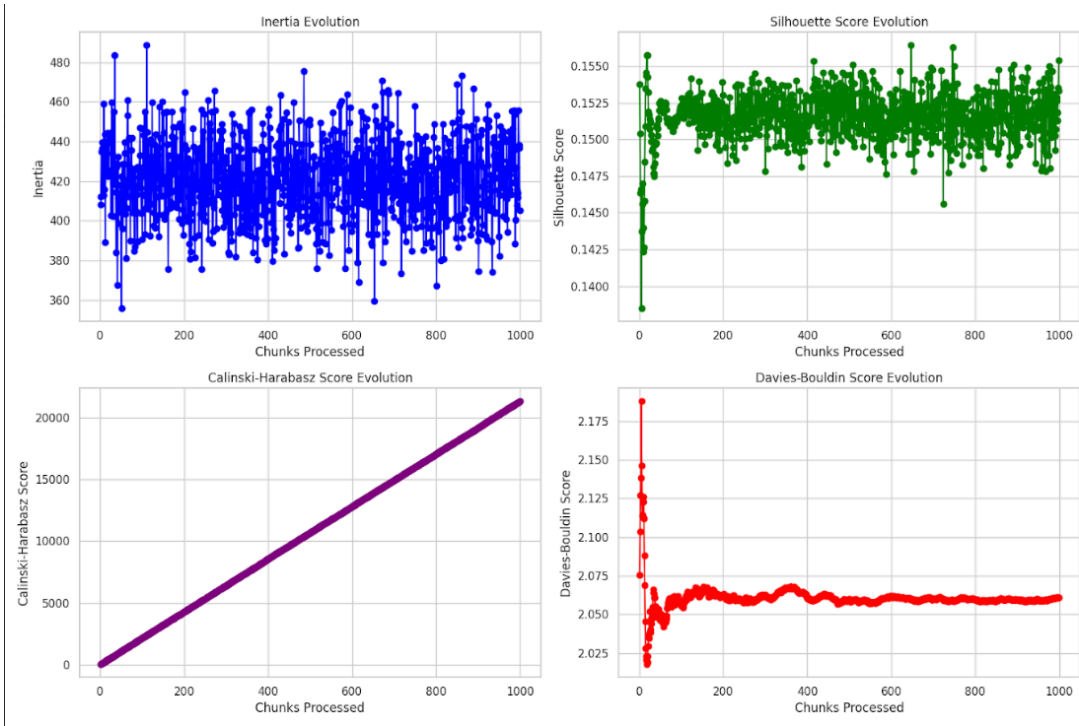


Feature selection

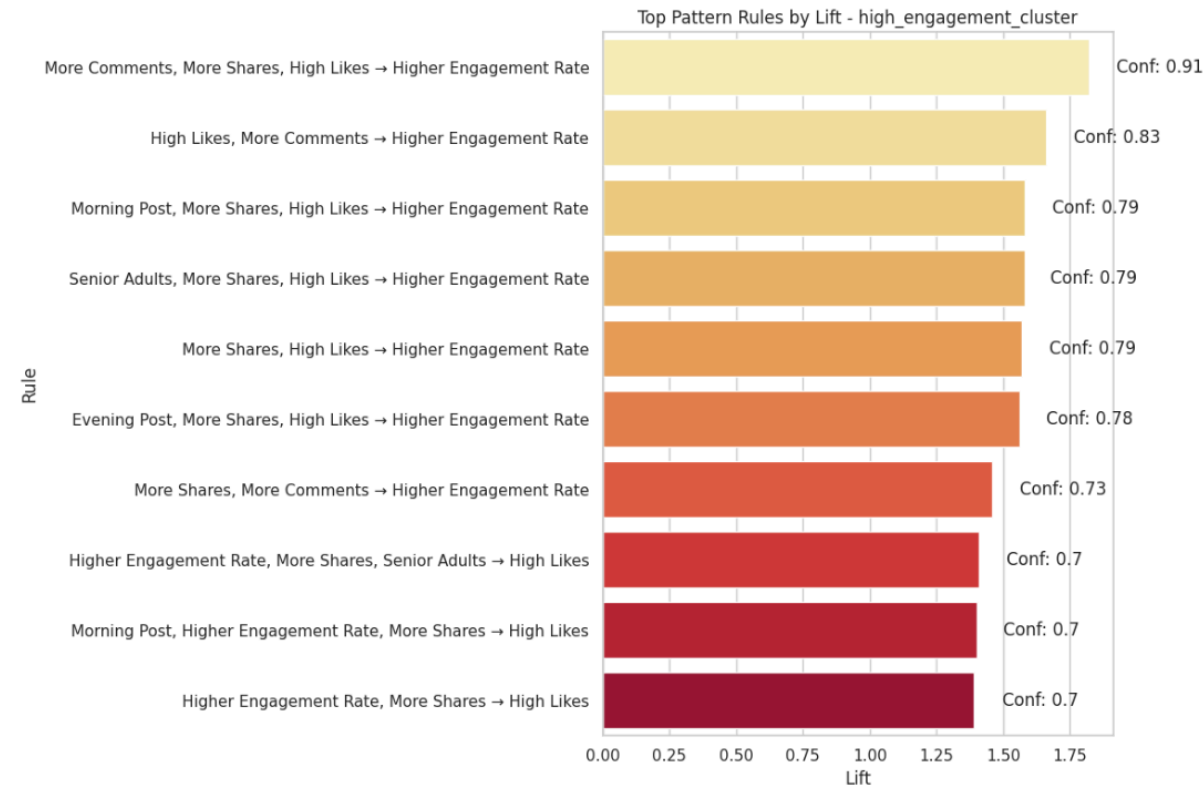




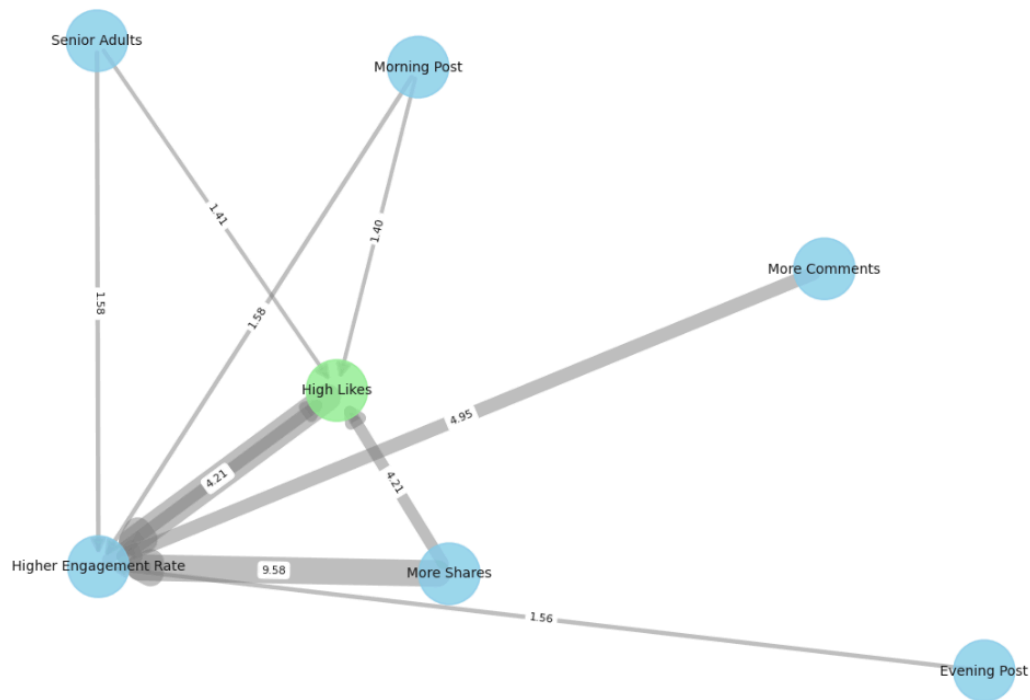
Evaluation/ performance measure:



Generalized Sequence Patterns Mining

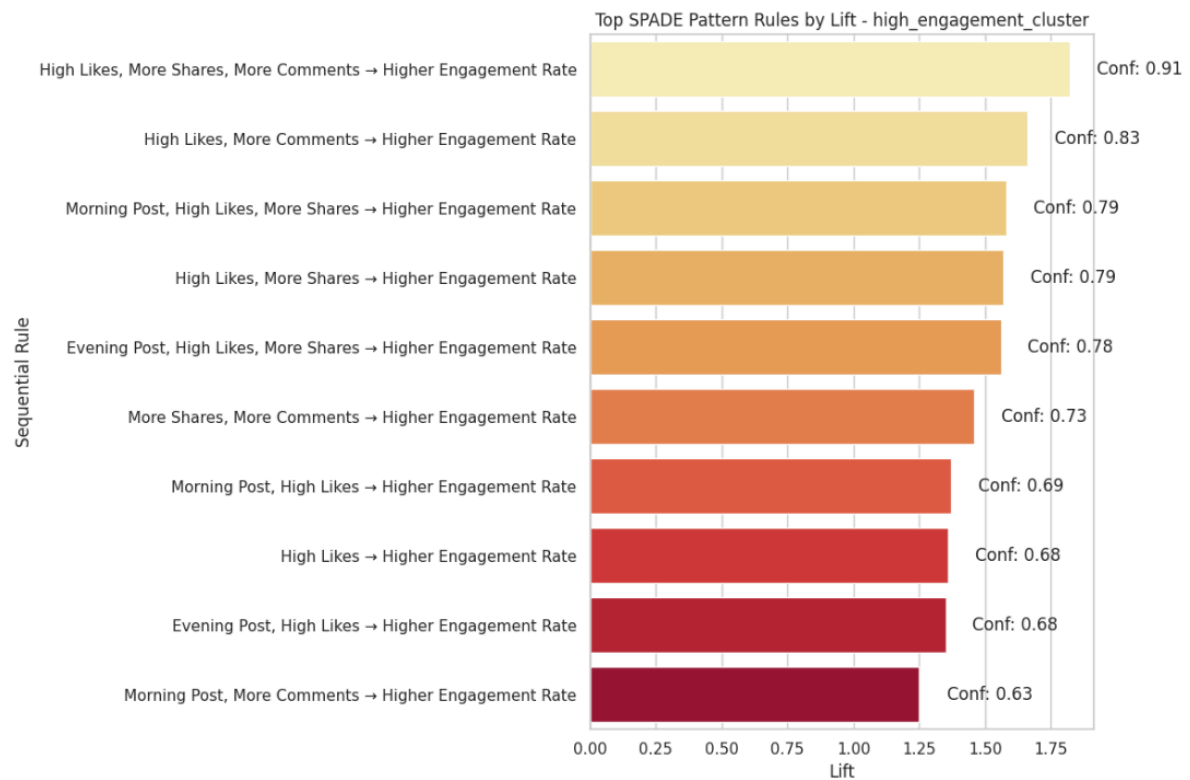


Pattern Sequence Analysis - high_engagement_cluster

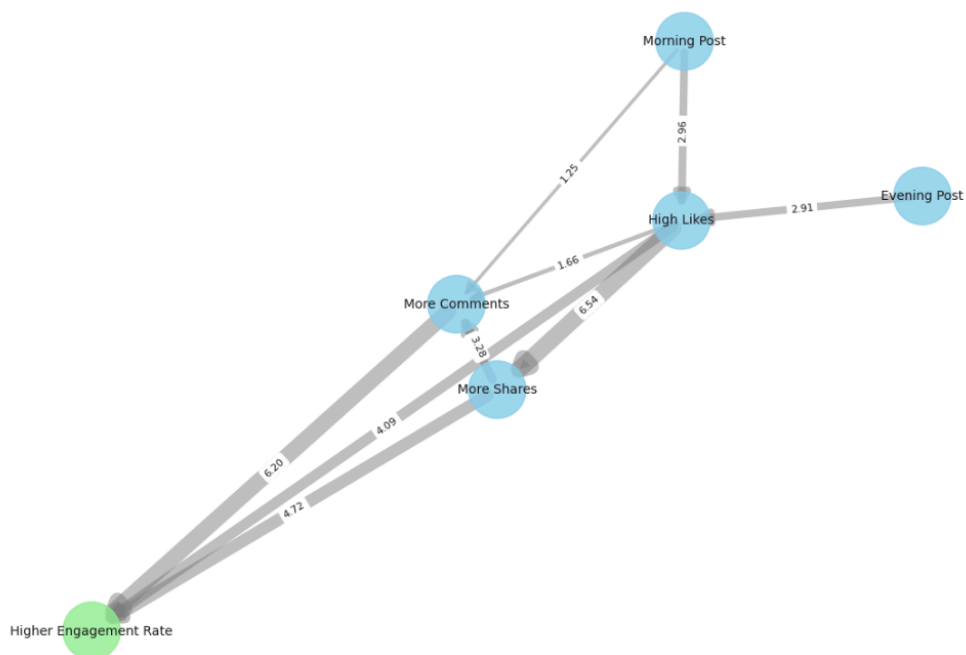


Antecedents	Consequents	Support	Confidence	Lift	Most Common Post Type	Most Common Gender
Senior Adults	More Comments	0.2401	0.5001	1.003	Video	Female
Senior Adults	Evening Post	0.2435	0.5073	1.0106	Link	Female
Senior Adults	High Likes	0.2419	0.5041	1.0112	Image	Female
Senior Adults	Higher Engagement Rate	0.2415	0.5031	1.0072	Video	Female
Senior Adults	More Shares	0.242	0.5041	1.0087	Image	Female
Morning Post	More Comments	0.2498	0.5015	1.0058	Video	Male
More Comments	Morning Post	0.2498	0.5009	1.0058	Video	Male
Higher Engagement Rate	More Comments	0.3098	0.6203	1.2439	Video	Female
More Comments	Higher Engagement Rate	0.3098	0.6214	1.2439	Video	Female
More Comments	More Shares	0.2494	0.5001	1.0006	Video	Female

SPADE

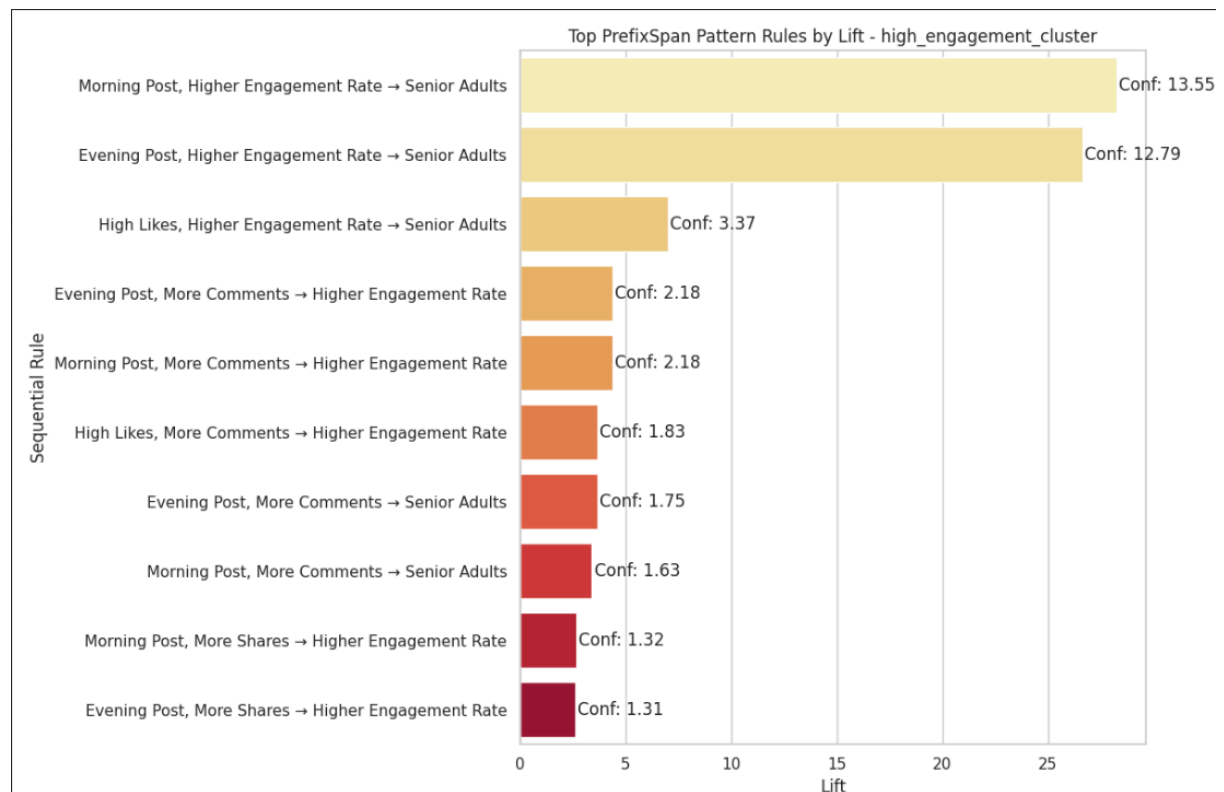


SPADE Pattern Sequence Analysis - high_engagement_cluster

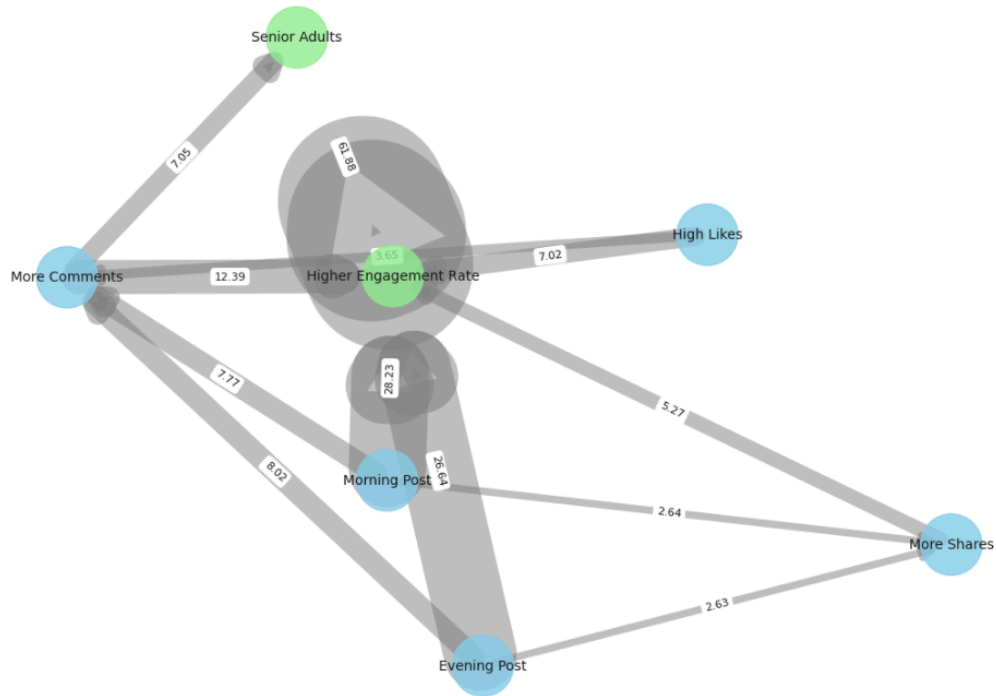


Antecedents	Consequents	Support	Confidence	Lift	Most Common Post Type	Most Common Gender
Evening Post	Higher Engagement Rate	0.2498	0.4976	0.9961	Link	Other
Evening Post, More Shares	Senior Adults	0.1239	0.4906	1.022	Link	Male
High Likes, More Comments	Senior Adults	0.1165	0.4937	1.0286	Image	Other
High Likes	Senior Adults	0.2419	0.4854	1.0112	Link	Female
Morning Post	High Likes	0.2476	0.4972	0.9975	Image	Male
High Likes, More Shares	Higher Engagement Rate	0.2113	0.7856	1.5727	Image	Other
Evening Post	Senior Adults	0.2435	0.4851	1.0106	Video	Male
More Shares, More Comments	Higher Engagement Rate	0.182	0.7299	1.4611	Video	Female
High Likes, More Shares	Senior Adults	0.1038	0.4915	1.0239	Image	Other
Higher Engagement Rate						
More Shares	Higher Engagement Rate	0.3039	0.6081	1.2173	Image	Male

PrefixSpan

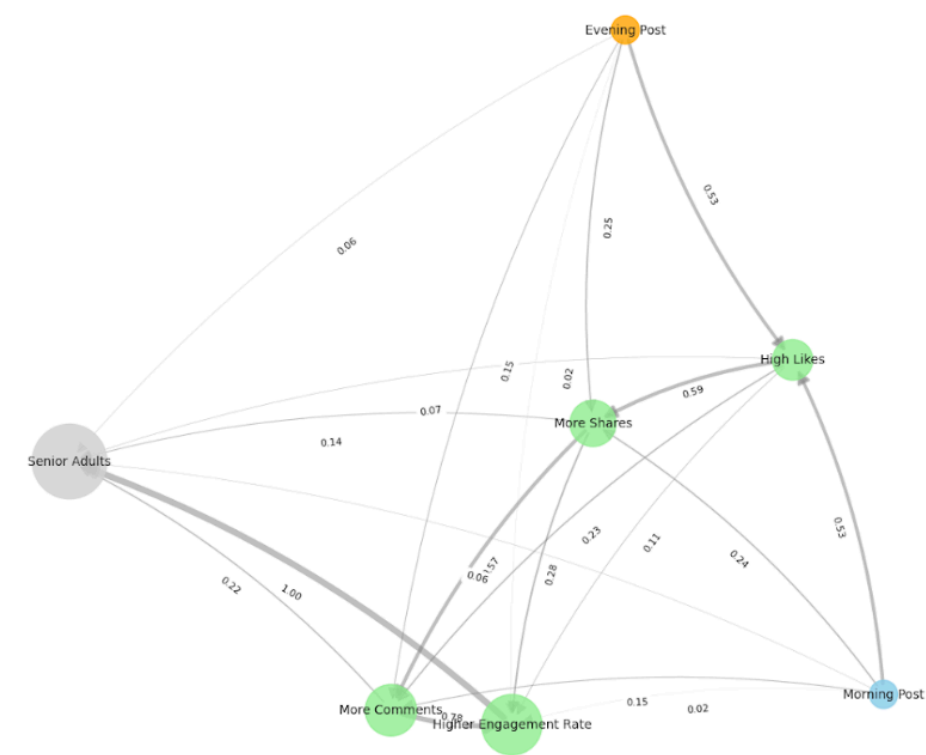


PrefixSpan Pattern Sequence Analysis - high_engagement_cluster



Antecedents	Consequents	Support	Confidence	Lift	Most Common Post Type	Most Common Gender
Morning Post	More Shares	0.2472	0.4963	0.9931	Link	Female
Morning Post, More Shares	Senior Adults	0.1181	1.033	2.152	Image	Female
Morning Post, More Shares	Higher Engagement Rate	0.1507	1.3189	2.6402	Video	Male
Morning Post, More Shares	More Comments	0.1235	1.0802	2.1662	Video	Other
Morning Post	Senior Adults	0.2365	0.4749	0.9893	Video	Female
Morning Post	High Likes	0.2476	0.4972	0.9975	Image	Male
Morning Post, High Likes	More Comments	0.1184	0.4782	0.959	Image	Female
Morning Post, High Likes	Higher Engagement Rate	0.17	0.6865	1.3743	Link	Male
Morning Post, High Likes	More Shares	0.1329	0.5366	1.0737	Image	Other
Morning Post, High Likes, More	Higher Engagement Rate	0.1052	0.7916	1.5848	Image	Other

Markov Chain State Transition Graph - high_engagement_cluster



Parallelization

